

Final Exam Review worksheet

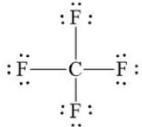
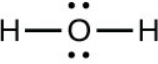
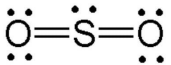
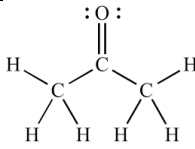
1. The following questions refer to the reversible reaction below:



It is observed that the equilibrium shifts to right when temperature is increased.

- ΔH for forward reaction: >0 or =0 or <0
- ΔS for forward reaction: >0 or =0 or <0
- In a coffee-cup calorimeter, the work will be done by or on the reaction when the reaction is moving to forward direction.
- This reaction will happen spontaneously under:
High temperature or low temperature
or any temperature or never happen spontaneously
- At equilibrium, this reaction prefers: reactant or product or neither.
- The reaction is at equilibrium and predict which direction the reaction will shift after each disturbance.
 - Adding NH_4HS into the reaction: shift left or shift right or remain unchanged.
 - Adding H_2S into the reaction: shift left or shift right or remain unchanged.
 - Decrease the volume of the reaction:
shift left or shift right or remain unchanged.
 - Increase the pressure by adding Ar gas in a rigid container:
shift left or shift right or remain unchanged.
- Initially, 0.356 mol of NH_4HS was placed in a 1.00 L sealed rigid container at 298 K. Calculate the concentration NH_3 , in M, at equilibrium at 298 K.

2. Determine if the system undergoes an exo- or endo- thermic process. The system is underlined.
- The evaporation of water from the skin:
 - The burning of candle wax:
 - The oxidation of iron in a chemical hand warmer:
 - The combustion of natural gas in a stove:
3. Determine polarity and intermolecular forces for the following compounds and their Lewis structures are given below:

Lewis structures	Polar?	Dispersion?	Dipole-Dipole?	H bond?	Ion-Dipole?
 CF ₄					
 H ₂ O					
 SO ₂					
 C ₃ H ₆ O					

Which compound is miscible with water (H₂O)?

CF₄

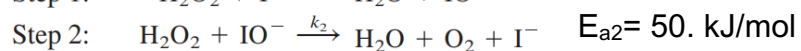
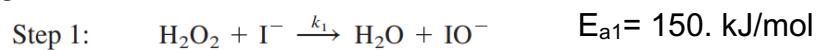
SO₂

C₃H₆O

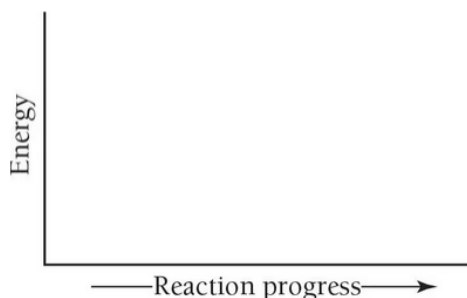
Which compound has the lowest melting point and why?

Which compound has the highest vapor pressure and why?

4. The following questions study a reaction mechanism, and the elementary steps are given below:

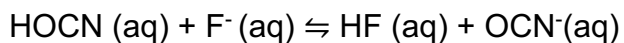


- a. The overall reaction has an enthalpy of -50. kJ/mol. Draw the reaction progress on the figure below and label E_{a1} , E_{a2} , and ΔH on the graph.



- b. Which is the rate determining step?
- c. Write the chemical equation for the overall reaction.
- d. Calculate the heat of the reaction when 3.52 g of H_2O_2 are reacted.
- e. The intermediate(s) in the reaction:
- f. The catalyst(s) in the reaction:
- g. Write the rate law for this reaction:
- h. If the H_2O_2 concentration is double, in what factor will the rate of overall reaction change?

5. Below is the Bronsted acid-base reaction:



a. Use the information on the information sheet to calculate the equilibrium constant for this reaction:

b. The weaker base for this reaction is:

c. At the beginning, the solution contains 0.15 M HOCN, 0.15 M F^- , 0.25 M HF, and 0.25 M OCN^- . Which direction will the reaction go?

6. Determine if the following salt is acid, neutral, or basic and rank the pH.

a. NaN_3

b. $\text{N}_2\text{H}_5\text{Cl}$

c. LiI

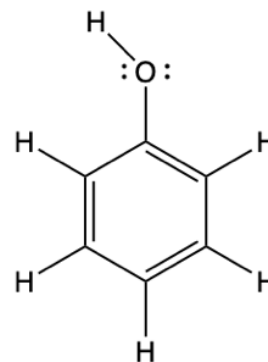
d. $\text{Mg}(\text{OCl})_2$

e. NH_4NO_3

7. On the right is the structure of an acid, phenol ($\text{HC}_6\text{H}_5\text{O}$).
Use the information on the information sheet and answer the questions below.

- a. Circle the H that is acidic.
b. Write the ionization reaction when phenol dissolves in water.

- c. Calculating pH and % ionization of the solution with 0.20 M of $\text{HC}_6\text{H}_5\text{O}$ at 25 °C.



- d. $\text{KC}_6\text{H}_5\text{O}$ solution is acidic or basic. Write the dissociation and ionization reaction when $\text{KC}_6\text{H}_5\text{O}$ dissolves in water.

- e. Calculating pH of the solution with 0.20 M of $\text{KC}_6\text{H}_5\text{O}$ at 25 °C.

- f. What is the best pH to target when $\text{HC}_6\text{H}_5\text{O}$ and $\text{KC}_6\text{H}_5\text{O}$ are used to make a buffer?
- g. A 1L buffer solution originally contains 0.20 M of $\text{HC}_6\text{H}_5\text{O}$ and 0.20 M of $\text{KC}_6\text{H}_5\text{O}$ at 25 °C. What will the pH be after 0.34 g of KOH is added into the buffer solution?

Answer: Q1g) 0.013M ; Q4d) -2.6 kJ; Q5a) 0.57; Q5c) backward; Q7c) 5.3 and 0.0025%;
Q7e) 11.6; Q7f) 9.89; Q7g) 9.94